

Red Hat OpenShift Data Foundation 4.22

Deploying OpenShift Data Foundation in external mode

Instructions for deploying OpenShift Data Foundation to use an external Red Hat Ceph Storage cluster and IBM FlashSystem.

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Red Hat OpenShift Data Foundation 4.22 Deploying OpenShift Data Foundation in external mode

Instructions for deploying OpenShift Data Foundation to use an external Red Hat Ceph Storage cluster and IBM FlashSystem.

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Abstract

Read this document for instructions on installing Red Hat OpenShift Data Foundation to use an external Red Hat Ceph Storage cluster or IBM FlashSystem.

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PREFACE

This document describes how to deploy Red Hat OpenShift Data Foundation in external mode, connecting to an existing Red Hat Ceph Storage cluster.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

CHAPTER 1. OVERVIEW OF DEPLOYING IN EXTERNAL MODE

Red Hat OpenShift Data Foundation can make services from an external Red Hat Ceph Storage cluster or use IBM FlashSystems that are available for consumption through OpenShift Container Platform clusters running on any platform. See the Planning your deployment guide for more information.

You can configure OpenShift Data Foundation in external mode to use an external IBM Storage Ceph cluster or IBM FlashSystem cluster using the steps provided in the following sections:

- Deploy OpenShift Data Foundation using Red Hat Ceph Storage.
- Deploy OpenShift Data Foundation using IBM FlashSystem.

For instructions regarding how to install a RHCS cluster, see the installation guide.

1.1. DISCONNECTED ENVIRONMENT REQUIREMENTS

For the list of packages to be mirrored for installing OpenShift Data Foundation in disconnected environments, see the Disconnected environment section in the *Planning your deployment* guide.

1.2. DISASTER RECOVERY REQUIREMENTS

Disaster Recovery features supported by Red Hat OpenShift Data Foundation require all of the following prerequisites to successfully implement a disaster recovery solution:

- A valid Red Hat OpenShift Data Foundation Advanced subscription
- A valid Red Hat Advanced Cluster Management for Kubernetes subscription

For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.

For detailed disaster recovery solution requirements, see the *Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads* guide, and the *Requirements and recommendations* section of the Install guide in Red Hat Advanced Cluster Management for Kubernetes documentation.

The following features related to external mode are not supported when using Metro-DR:

- StorageClasses using non-default RADOS namespace
- User created StorageClasses, even when using default RADOS namespace
- Multiple StorageClasses

1.3. NETWORK PORTS REQUIRED BETWEEN OPENSHIFT CONTAINER PLATFORM AND CEPH WHEN USING EXTERNAL MODE DEPLOYMENT

For detailed information about the network ports required between OpenShift Container Platform and Ceph when using external mode deployment, see the Network ports for external mode section in the OpenShift Data Foundation Architecture guide.

Additional resources

- Planning your deployment

- Disconnected environment
- Deploy OpenShift Data Foundation using Red Hat Ceph Storage
- Deploy OpenShift Data Foundation using IBM FlashSystem
- Red Hat Ceph Storage installation guide
- Knowledgebase article on OpenShift Data Foundation subscriptions
- Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads
- Red Hat Advanced Cluster Management for Kubernetes - Requirements and recommendations
- Network ports for external mode

CHAPTER 2. DEPLOY OPENSHIFT DATA FOUNDATION USING RED HAT CEPH STORAGE

Red Hat OpenShift Data Foundation can make services from an external Red Hat Ceph Storage cluster available for consumption through OpenShift Container Platform clusters. You need to install the OpenShift Data Foundation operator and then create an OpenShift Data Foundation cluster for the external Ceph storage system.

2.1. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
   d. Select Approval Strategy as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.

e. Ensure that the **Enable** option is selected for the **Console plugin**.

f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
  - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
  - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

2.2. CREATING AN OPENSHIFT DATA FOUNDATION CLUSTER FOR EXTERNAL CEPH STORAGE SYSTEM

Create an OpenShift Data Foundation cluster service in external mode by importing the Ceph cluster connection details, which enables OpenShift workloads to consume block, file, and object storage from the external Red Hat Ceph Storage cluster.

You need to create a new OpenShift Data Foundation cluster after you install OpenShift Data Foundation operator on OpenShift Container Platform deployed on VMware vSphere or user-provisioned bare metal infrastructures.

Prerequisites

- A valid Red Hat OpenShift Data Foundation Advanced subscription. To know how subscriptions for OpenShift Data Foundation work, see knowledgebase article on OpenShift Data Foundation subscriptions.
- Ensure the OpenShift Container Platform version is 4.22 or above before deploying OpenShift Data Foundation 4.22.
- OpenShift Data Foundation operator must be installed. For more information, see Installing OpenShift Data Foundation Operator using the Software Catalog.
- To check the supportability and interoperability of Red Hat Ceph Storage (RHCS) with Red Hat OpenShift Data Foundation in external mode, go to the lab Red Hat OpenShift Data Foundation Supportability and Interoperability Checker.
  - Select **Service Type** as **ODF as Self-Managed Service**.
  - Select appropriate **Version** from the drop down.
  - On the Versions tab, click the **Supported RHCS versions in the External Mode**tab.
- Red Hat Ceph Storage must have Ceph Dashboard installed and configured. For more information, see Ceph Dashboard installation and access .

- It is recommended that the external Red Hat Ceph Storage cluster has the PG Autoscaler enabled. For more information, see The placement group autoscaler section in the Red Hat Ceph Storage documentation.

Procedure

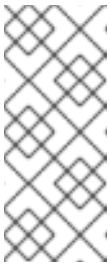
1. In the OpenShift Web Console, click **Storage → External Systems**.
2. Click **Connect external systems**.
3. Select **Red Hat/IBM Ceph Cluster** and click **Next**.
4. In the Connect Red Hat Ceph storage page, provide the necessary information:
   a. Click on the **Download Script** link to download the python script for extracting Ceph cluster details, or extract it using command line:

```
$ oc get cm rook-ceph-external-cluster-script-config -n openshift-storage -o
jsonpath='{.data.script}' | base64 --decode > ceph-external-cluster-details-exporter.py
```

   b. For extracting the Red Hat Ceph Storage (RHCS) cluster details, contact the RHCS administrator to run the downloaded python script on a Red Hat Ceph Storage node with the **admin key**.
      i. Run the following command on the RHCS node to view the list of available arguments:

```
# python3 ceph-external-cluster-details-exporter.py --help
```

      You can also run the script from inside a MON container (containerized deployment) or from a MON node (RPM deployment).



NOTE

Use the **yum install cephadm** command and then the **cephadm** command to deploy your RHCS cluster using containers. You must pull the RHCS container images using the **cephadm** command, rather than using **yum** for installing the Ceph packages onto nodes. For more information, see RHCS product documentation.

      ii. To retrieve the external cluster details from the RHCS cluster, choose one of the following two options, either a configuration file or command-line flags.

      A. **Configuration file**
      Use **config-file** flag. This stores the parameters used during deployment.

      In new deployments, you can save the parameters used during deployment in a configuration file. This file can then be used during upgrade to preserve the parameters as well as adding any additional parameters. Use **config-file** to set the path to the configuration file.

      Example configuration file saved in /**config.ini**:

```
[Configurations]
format = bash
cephfs-filesystem-name = <filesystem-name>
```

```
rbd-data-pool-name = <pool_name>
...
```

Set the path to the **config.ini** file using **config-file**:

```
# python3 ceph-external-cluster-details-exporter.py --config-file /config.ini
```

B. **Command-line flags**
Retrieve the external cluster details from the RHCS cluster, and pass the parameters for your deployment.

```
# python3 ceph-external-cluster-details-exporter.py \
--rbd-data-pool-name <rbd block pool name>  [optional arguments]
```

For example:

```
# python3 ceph-external-cluster-details-exporter.py --rbd-data-pool-name ceph-rbd --monitoring-endpoint xxx.xxx.xxx.xxx --monitoring-endpoint-port xxxx --rgw-endpoint xxx.xxx.xxx.xxx:xxxx --run-as-user client.ocs
```

RBD parameters

rbd-data-pool-name

A mandatory parameter that is used for providing block storage in OpenShift Data Foundation.

rados-namespace

Divides an RBD data pool into separate logical namespaces, used for creating RBD PVC in a **radosNamespace**. Flags required with **rados-namespace** are **restricted-auth-permission** and **k8s-cluster-name**.

rbd-metadata-ec-pool-name

(Optional) The name of the erasure coded RBD metadata pool.

RGW parameters

rgw-endpoint

(Optional) This parameter is required only if the object storage is to be provisioned through Ceph Rados Gateway for OpenShift Data Foundation. Provide the endpoint in the following format: **<ip_address>:<port>**



NOTE

A fully-qualified domain name (FQDN) is also supported in the format **<FQDN>:<PORT>**.

rgw-pool-prefix

(Optional) The prefix of the RGW pools. If not specified, the default prefix is **default**.

rgw-tls-cert-path

(Optional) The file path of the RADOS Gateway endpoint TLS certificate.

- To provide the TLS certificate and RGW endpoint details to the helper script, **ceph-external-cluster-details-exporter.py**, run the following command:

```
# python3 ceph-external-clustergw-endpoint r-details-exporter.py --rbd-data-pool-name <rbd block pool name> --rgw-endpoint <ip_address>:<port> --rgw-tls-cert-path <file path containing cert>
```

This creates a resource to create a Ceph Object Store CR such as Kubernetes Secret containing the TLS certificate. All the intermediate certificates including private keys need to be stored in the certificate file.

rgw-skip-tls

(Optional) This parameter ignores the TLS certification validation when a self-signed certificate is provided (not recommended).

Monitoring parameters

monitoring-endpoint

(Optional) This parameter accepts comma-separated list of IP addresses of active and standby **mgrs** reachable from the OpenShift Container Platform cluster. If not provided, the value is automatically populated.

monitoring-endpoint-port

(Optional) It is the port associated with the **ceph-mgr** Prometheus exporter specified by **--monitoring-endpoint**. If not provided, the value is automatically populated.

Ceph parameters

ceph-conf

(Optional) The name of the Ceph configuration file.

run-as-user

(Optional) This parameter is used for providing name for the Ceph user which is created by the script. If this parameter is not specified, a default user name **client.healthchecker** is created. The permissions for the new user is set as:

- caps: [mgr] allow command config
- caps: [mon] allow r, allow command quorum_status, allow command version
- caps: [osd] allow rwx pool=**RGW_POOL_PREFIX.rgw.meta**, allow r pool=**.rgw.root**, allow rw pool=**RGW_POOL_PREFIX.rgw.control**, allow rx pool=**RGW_POOL_PREFIX.rgw.log**, allow x pool=**RGW_POOL_PREFIX.rgw.buckets.index**

CephFS parameters

cephfs-metadata-pool-name

(Optional) The name of the CephFS metadata pool.

cephfs-data-pool-name

(Optional) The name of the CephFS data pool.

cephfs-filesystem-name

(Optional) The name of the CephFS filesystem.

Output parameters

dry-run

(Optional) This parameter helps to print the executed commands without running them.

output

(Optional) The file where the output is required to be stored.

Multicluster parameters

k8s-cluster-name

(Optional) Kubernetes cluster name.

cluster-name

(Optional) The Ceph cluster name.

restricted-auth-permission

(Optional) This parameter restricts **cephCSIKeyrings** auth permissions to specific pools and clusters. Mandatory flags that need to be set with this are **rbd-data-pool-name** and **cluster-name**. You can also pass the **cephfs-filesystem-name** flag if there is CephFS user restriction so that permission is restricted to a particular CephFS filesystem.



NOTE

This parameter must be applied only for the new deployments. To restrict **csi-users** per pool and per cluster, you need to create new **csi-users** and new secrets for those **csi-users**.

Example with restricted auth permission:

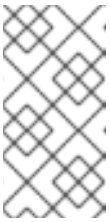
```
# python3 /etc/ceph/create-external-cluster-resources.py --cephfs-filesystem-name myfs --rbd-data-pool-name replicapool --cluster-name rookStorage --restricted-auth-permission true
```

Example of JSON output generated using the python script:

[{"name": "rook-ceph-mon-endpoints", "kind": "ConfigMap", "data": {"data": "xxx.xxx.xxx.xxx:xxxx", "maxMonId": "0", "mapping": "{}"}}, {"name": "rook-ceph-mon", "kind": "Secret", "data": {"admin-secret": "admin-secret", "fsid": "<fs-id>", "mon-secret": "mon-secret"}}, {"name": "rook-ceph-operator-creds", "kind": "Secret", "data": {"userID": "<user-id>", "userKey": "<user-key>"}}, {"name": "rook-csi-rbd-node", "kind": "Secret", "data": {"userID": "csi-rbd-node", "userKey": "<user-key>"}}, {"name": "ceph-rbd", "kind": "StorageClass", "data": {"pool": "<pool>"}}, {"name": "monitoring-endpoint", "kind": "CephCluster", "data": {"MonitoringEndpoint": "xxx.xxx.xxx.xxx", "MonitoringPort": "xxxx"}}, {"name": "rook-ceph-dashboard-link", "kind": "Secret", "data": {"userID": "ceph-dashboard-link", "userKey": "<user-key>"}}, {"name": "rook-csi-rbd-provisioner", "kind": "Secret", "data": {"userID": "csi-rbd-provisioner", "userKey": "<user-key>"}}, {"name": "rook-csi-cephfs-provisioner", "kind": "Secret", "data": {"adminID": "csi-cephfs-provisioner", "adminKey": "<admin-key>"}}, {"name": "rook-csi-cephfs-node", "kind": "Secret", "data": {"adminID": "csi-cephfs-node", "adminKey": "

```
<admin-key>"}}, {"name": "cephfs", "kind": "StorageClass", "data": {"fsName": "cephfs",
"pool": "cephfs_data"}}, {"name": "ceph-rgw", "kind": "StorageClass", "data": {"endpoint":
"xxx.xxx.xxx.xxx:xxxx", "poolPrefix": "default"}}, {"name": "rgw-admin-ops-user", "kind":
"Secret", "data": {"accessKey": "<access-key>", "secretKey": "<secret-key>"}}]
```

i. Save the JSON output to a file with **.json** extension



NOTE

For OpenShift Data Foundation to work seamlessly, ensure that the parameters (RGW endpoint, CephFS details, RBD pool, and so on) to be uploaded using the JSON file remain unchanged on the RHCS external cluster after the storage cluster creation.

ii. Run the command when there is a multi-tenant deployment in which the RHCS cluster is already connected to OpenShift Data Foundation deployment with a lower version.

```
# python3 ceph-external-cluster-details-exporter.py --upgrade
```

a. Click **Browse** to select and upload the JSON file.
The content of the JSON file is populated and displayed in the text box.

b. (Optional) Select **In-transit encryption** to enable encryption for entire Ceph traffic including data using Ceh **msgrv2**.



NOTE

Verify that the external Red Hat Ceph Storage (RHCS) cluster is configured for in-transit encryption if required. Follow the documentation for detailed configuration steps before enabling in-transit encryption.

c. (Optional) Select the advanced configuration options that apply to your environment:

iii. **Use Ceph RBD as the default StorageClass**
Sets the RBD StorageClass as the cluster default.

iv. **Use external PostgreSQL** *(Technology preview)*
Uses an external PostgreSQL instance instead of the in-cluster database.

Additional advanced options may be available in the UI. Select only those that are required for the deployment policy and environment.

v. **Enable automatic backup**
Allows automatic backup for the Multicloud Object Gateway metadata database.

A. Select the backup frequency: Daily, Weekly, or Monthly.

B. Specify the number of backups to retain.



NOTE

This option is disabled if the **Use external PostgreSQL** option is selected.

5. Review the selections and click **Connect**.

Verification

To verify the final Status of the installed storage cluster:

1. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System → ocs-external-storagecluster**.
2. Verify that **Status** of **StorageCluster** is **Ready** and has a green tick.
3. To verify that OpenShift Data Foundation, pods and StorageClass are successfully installed, see Verifying your external mode OpenShift Data Foundation installation for external Ceph storage system.

2.3. CONFIGURING TOPOLOGY AWARENESS FOR EXTERNAL MODE

Topology awareness ensures that data for persistent volumes (PVs) is provisioned from Ceph OSDs that belong to the same failure domain as the OpenShift node where the application pod runs.

Prerequisites

- You have installed OpenShift Data Foundation in external mode.
- You have access to the external Red Hat Ceph Storage cluster with administrative privileges.
- You know the Ceph failure domain that you want to use for topology-aware provisioning.
- The OpenShift worker nodes that host the application workloads have labels that map to the Ceph failure domain values.

Procedure

1. Identify the Ceph failure domain and the corresponding OpenShift nodes.
   a. Check the Ceph failure domain:

   ```
   $ ceph osd tree
   ```

   Example output:

   ```
   ID  CLASS  WEIGHT   TYPE NAME      STATUS  REWEIGHT  PRI-AFF
   ...
   -9         0.50000      host osd-0
    3    hdd  0.25000          osd.3      up   1.00000  1.00000
   ...
   ```

   In this example, the failure domain is **host**, and the device class is **hdd**.

   b. Identify the OpenShift worker nodes and map them to the Ceph failure domains:

   ```
   $ oc get nodes
   ```

   Example output:

```
NAME          STATUS   ROLES    AGE    VERSION
compute-0      Ready    worker   2d23h   v1.34.1
compute-1      Ready    worker   2d23h   v1.34.1
compute-2      Ready    worker   2d23h   v1.34.1
```

c. Verify the node labels that you want to use for topology awareness:

```
$ oc get nodes --show-labels
```

In this example, the label **topology.kubernetes.io/hostname** and its values are chosen, for example, **compute-0**, **compute-1**, and **compute-2**, when configuring topology-aware external cluster resources.

2. Create separate block pools for each failure domain.
In this example, a replica-3 pool is created for each failure domain.

```
$ ceph osd pool create pool-1-data 32 32 replicated
$ ceph osd pool set pool-1-data size 3
$ ceph osd pool create pool-2-data 32 32 replicated
$ ceph osd pool set pool-2-data size 3
$ ceph osd pool create pool-3-data 32 32 replicated
$ ceph osd pool set pool-3-data size 3
```

Verify the pools:

```
$ ceph osd pool ls detail
```

3. Generate the external cluster resources with topology-aware settings.
Run the following command on the Ceph client:

```
$ python3 /etc/rook-external/create-external-cluster-resources.py \
  --rbd-data-pool-name rbd \
  --topology-pools pool-1-data,pool-2-data,pool-3-data \
  --topology-failure-domain-label host \
  --topology-failure-domain-values compute-0,compute-1,compute-2
```

where

--rbd-data-pool-name

Specifies an existing RBD pool.

--topology-pools

Specifies the topology-constrained pools created for each failure domain.

--topology-failure-domain-label

Specifies the Ceph failure domain label, for example, **host**.

--topology-failure-domain-values

Specifies the corresponding OpenShift failure domain values, for example, **compute-0,compute-1,compute-2**.

4. Upload the generated JSON file when creating the external cluster in the OpenShift web console.

5. Verify that topology awareness is configured.

   a. Verify that the operator configuration contains the topology domain label:

   ```
   $ oc get operatorconfigs.csi.ceph.io -o yaml | grep topology -A3
   ```

   Expected output:

   ```
   topology:
     domainLabels:
     - kubernetes.io/hostname
   ```

   b. Verify that the topology-aware storage class contains the topology parameters:

   ```
   $ oc get sc ocs-external-storagecluster-ceph-non-resilient-rbd -o yaml
   ```

   Example output:

   ```
   parameters:
     topologyConstrainedPools: |-
       [
         {
           "poolName": "pool-1-data",
           "domainSegments": [
             {
               "domainLabel": "hostname",
               "value": "compute-0"
             }
           ]
         },
         {
           "poolName": "pool-2-data",
           "domainSegments": [
             {
               "domainLabel": "hostname",
               "value": "compute-1"
             }
           ]
         },
         {
           "poolName": "pool-3-data",
           "domainSegments": [
             {
               "domainLabel": "hostname",
               "value": "compute-2"
             }
           ]
         }
       ]
     topologyFailureDomainLabel: host
     topologyFailureDomainValues: compute-0,compute-1,compute-2
     topologyPools: pool-1-data,pool-2-data,pool-3-data
   ```

6. Test topology-aware provisioning by creating a persistent volume claim (PVC) and a pod. Apply the following YAML:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: topo-verify-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: ocs-external-storagecluster-ceph-non-resilient-rbd
---
apiVersion: v1
kind: Pod
metadata:
  name: topo-verify-pod
spec:
  containers:
  - name: test-container
    image: busybox
    command: [ "sh", "-c", "sleep 3600" ]
    volumeMounts:
    - mountPath: /data
      name: my-volume
  volumes:
  - name: my-volume
    persistentVolumeClaim:
      claimName: topo-verify-pvc
```

The **WaitForFirstConsumer** volume binding mode ensures that the PV is provisioned only after the pod is scheduled, so topology constraints can be applied.

7. Verify data placement in the expected failure domain.

   a. Get the image name for the provisioned PV:

   ```
   $ kubectl get pv $(kubectl get pvc topo-verify-pvc -o jsonpath='{.spec.volumeName}') -o
   jsonpath='{.spec.csi.volumeAttributes.imageName}'
   ```

   b. Check usage from the Ceph side:

   ```
   $ rbd du pool-1-data/<image_name>
   ```

   c. Identify the OSDs that store the image data:

   ```
   $ ceph osd map pool-1-data rbd_header.<image_name>
   ```

   Example output:

   ```
   ... -> pg 14.1ed52b0 (14.10) -> up ([4,5,6], p4) acting ([4,5,6], p4)
   ```

   d. Verify that the OSDs belong to the expected failure domains:

   ```
   $ ceph osd tree | grep -E "osd.4|osd.5|osd.6"
   ```

Example output:

```
5   hdd  0.25000        osd.5      up   1.00000  1.00000
6   hdd  0.25000        osd.6      up   1.00000  1.00000
4   hdd  0.25000        osd.4      up   1.00000  1.00000
```

2.4. VERIFYING YOUR OPENSHIFT DATA FOUNDATION INSTALLATION FOR EXTERNAL CEPH STORAGE SYSTEM

Verify your OpenShift Data Foundation external mode installation to confirm that the storage cluster is healthy, the expected storage classes are present, and application pods can successfully provision and use PVCs.

2.4.1. Verifying the state of the pods

1. Click **Workloads → Pods** from the left pane of the OpenShift Web Console.
2. Select **openshift-storage** from the **Project** drop-down list.



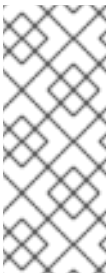
NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

3. Verify that the following pods are in running state:
   Pods corresponding to OpenShift Data Foundation components

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any worker node) • **ocs-metrics-exporter-*** (1 pod on any worker node) • **odf-operator-controller-manager-*** (1 pod on any worker node) • **odf-console-*** (1 pod on any worker node) • **csi-addons-controller-manager-*** (1 pod on any worker node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any worker node)

Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any worker node) • **noobaa-core-*** (1 pod on any worker node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)

CSI	• **cephfs** ◦ **openshift-storage.cephfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node)  NOTE If an MDS is not deployed in the external cluster, the **openshift-storage.cephfs.csi.ceph.com-ctrlplugin** and **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** pods are not created. • **nfs** ◦ **openshift-storage.nfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.nfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **rbd** ◦ **openshift-storage.rbd.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.rbd.csi.ceph.com-nodeplugin-*** (1 pod on each storage node)

2.4.2. Verifying that the OpenShift Data Foundation cluster is healthy

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. In the **Status** card of the **Block and File** tab, verify that the *Storage Cluster* has a green tick.
4. In the **Details** card, verify that the cluster information is displayed.

2.4.3. Verifying that the Multicloud Object Gateway is healthy

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
   a. In the **Status** card of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.
   b. In the **Details** card, verify that the Multicloud Object Gateway (MCG) information is displayed.



NOTE

The RADOS Object Gateway is only listed in case RADOS Object Gateway endpoint details are included while deploying OpenShift Data Foundation in external mode.

2.4.4. Verifying that the storage classes are created and listed

1. Click **Storage → Storage Classes**from the left pane of the OpenShift Web Console.
2. Verify that the following storage classes are created with the OpenShift Data Foundation cluster creation:
   - **ocs-external-storagecluster-ceph-rbd**
   - **ocs-external-storagecluster-ceph-rgw**
   - **ocs-external-storagecluster-cephfs**
   - **openshift-storage.noobaa.io**



NOTE

- If an MDS is not deployed in the external cluster, **ocs-external-storagecluster-cephfs** storage class will not be created.
- If RGW is not deployed in the external cluster, the **ocs-external-storagecluster-ceph-rgw** storage class will not be created.

For more information regarding MDS and RGW, see the Red Hat Ceph Storage documentation.

Additional resources

- Monitoring OpenShift Data Foundation
- Red Hat Ceph Storage documentation

2.4.5. Verifying that Ceph cluster is connected

Run the following command to verify if the OpenShift Data Foundation cluster is connected to the external Red Hat Ceph Storage cluster.

```
$ oc get cephcluster -n openshift-storage
NAME                                DATADIRHOSTPATH   MONCOUNT   AGE   PHASE       MESSAGE
HEALTH      EXTERNAL
ocs-external-storagecluster-cephcluster                        30m   Connected   Cluster connected
successfully   HEALTH_OK   true
```

2.4.6. Verifying that storage cluster is ready

Run the following command to verify if the storage cluster is ready and the **External** option is set to **true**.

```
$ oc get storagecluster -n openshift-storage
NAME                          AGE   PHASE   EXTERNAL   CREATED AT            VERSION
ocs-external-storagecluster   30m   Ready   true       2021-11-17T09:09:52Z   /4.22/.0
```

2.4.7. Verifying the the creation of Ceph Object Store CRD

Run the following command to verify if the Ceph Object Store CRD is created in the external Red Hat Ceph Storage cluster.

```
$ oc get cephobjectstore -n openshift-storage"
NAME              PHASE     ENDPOINT              SECUREENDPOINT      AGE
object-store1      Ready     <http://IP/FQDN:port>                    15m
```

2.4.8. Verifying the state of the pods

Verify that all expected OpenShift Data Foundation pods are running in the **openshift-storage** namespace after deploying in external mode.

Procedure

1. Click **Workloads → Pods** from the left pane of the OpenShift Web Console.
2. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

3. Verify that the following pods are in running state:
   Pods corresponding to OpenShift Data Foundation components

Component	Corresponding pods

OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any worker node) • **ocs-metrics-exporter-*** (1 pod on any worker node) • **odf-operator-controller-manager-*** (1 pod on any worker node) • **odf-console-*** (1 pod on any worker node) • **csi-addons-controller-manager-*** (1 pod on any worker node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any worker node)
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any worker node) • **noobaa-core-*** (1 pod on any worker node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)

CSI	• **cephfs** ◦ **openshift-storage.cephfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) NOTE If an MDS is not deployed in the external cluster, the **openshift-storage.cephfs.csi.ceph.com-ctrlplugin** and **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** pods are not created. • **nfs** ◦ **openshift-storage.nfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.nfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **rbd** ◦ **openshift-storage.rbd.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.rbd.csi.ceph.com-nodeplugin-*** (1 pod on each storage node)

2.4.9. Verifying that the OpenShift Data Foundation cluster is healthy

Verify the health status of the OpenShift Data Foundation storage cluster in the Block and File tab of the OpenShift web console after deploying in external mode.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.

3. In the **Status** card of the **Block and File** tab, verify that the *Storage Cluster* has a green tick.

4. In the **Details** card, verify that the cluster information is displayed.
   For more information on the health of OpenShift Data Foundation cluster using the Block and File dashboard, see the Monitoring OpenShift Data Foundation documentation.

Additional resources

- Monitoring OpenShift Data Foundation

2.4.10. Verifying that the Multicloud Object Gateway is healthy

Verify that the Multicloud Object Gateway is healthy and both Object Service and Data Resiliency show green status after deploying OpenShift Data Foundation in external mode.

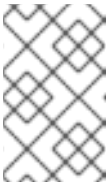
Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.

2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.

   a. In the **Status** card of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.

   b. In the **Details** card, verify that the Multicloud Object Gateway (MCG) information is displayed.

   NOTE

   The RADOS Object Gateway is only listed in case RADOS Object Gateway endpoint details are included while deploying OpenShift Data Foundation in external mode.

   For more information on the health of OpenShift Data Foundation cluster using the object dashboard, see the Monitoring OpenShift Data Foundation documentation.

Additional resources

- Monitoring OpenShift Data Foundation

2.4.11. Verifying that the storage classes are created and listed

Verify that the expected OpenShift Data Foundation storage classes are created and listed after deploying in external mode.

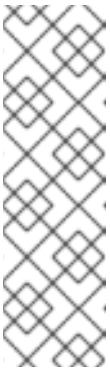
Procedure

1. Click **Storage → Storage Classes**from the left pane of the OpenShift Web Console.

2. Verify that the following storage classes are created with the OpenShift Data Foundation cluster creation:

   - **ocs-external-storagecluster-ceph-rbd**

- **ocs-external-storagecluster-ceph-rgw**
- **ocs-external-storagecluster-cephfs**
- **openshift-storage.noobaa.io**



NOTE

- If an MDS is not deployed in the external cluster, **ocs-external-storagecluster-cephfs** storage class will not be created.
- If RGW is not deployed in the external cluster, the **ocs-external-storagecluster-ceph-rgw** storage class will not be created.

For more information regarding MDS and RGW, see the Red Hat Ceph Storage documentation.

Additional resources

- Red Hat Ceph Storage documentation

2.4.12. Verifying that Ceph cluster is connected

Verify that the OpenShift Data Foundation cluster is connected to the external Red Hat Ceph Storage cluster after deploying in external mode.

Procedure

1. Run the following command to verify if the OpenShift Data Foundation cluster is connected to the external Red Hat Ceph Storage cluster:

```
$ oc get cephcluster -n openshift-storage
NAME                                DATADIRHOSTPATH   MONCOUNT   AGE   PHASE
MESSAGE                  HEALTH      EXTERNAL
ocs-external-storagecluster-cephcluster                        30m   Connected   Cluster
connected successfully   HEALTH_OK   true
```

2.4.13. Verifying that storage cluster is ready

Verify that the external OpenShift Data Foundation storage cluster is ready and the **External** flag is set to true after deployment.

Procedure

1. Run the following command to verify if the storage cluster is ready and the **External** option is set to **true**:

```
$ oc get storagecluster -n openshift-storage
NAME                          AGE   PHASE   EXTERNAL   CREATED AT            VERSION
ocs-external-storagecluster   30m   Ready   true       2021-11-17T09:09:52Z   /4.22/.0
```

2.4.14. Verifying the creation of Ceph Object Store CRD

Verify that the Ceph Object Store CRD has been created in the external Red Hat Ceph Storage cluster after deploying OpenShift Data Foundation in external mode.

Procedure

1. Run the following command to verify if the Ceph Object Store CRD is created in the external Red Hat Ceph Storage cluster:

```
$ oc get cephobjectstore -n openshift-storage
NAME             PHASE   ENDPOINT                SECUREENDPOINT   AGE
object-store1    Ready   <http://IP/FQDN:port>                    15m
```

CHAPTER 3. DEPLOY OPENSHIFT DATA FOUNDATION USING IBM FLASHSYSTEM

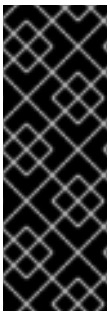
OpenShift Data Foundation can use IBM FlashSystem storage available for consumption through OpenShift Container Platform clusters. You need to install the OpenShift Data Foundation operator and then create an OpenShift Data Foundation cluster for IBM FlashSystem storage.

3.1. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**.
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
   d. Select Approval Strategy as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.

e. Ensure that the **Enable** option is selected for the **Console plugin**.

f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
  - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
  - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

3.2. CREATING AN OPENSHIFT DATA FOUNDATION CLUSTER FOR EXTERNAL IBM FLASHSYSTEM STORAGE

You need to create a new OpenShift Data Foundation cluster after you install the OpenShift Data Foundation operator on the OpenShift Container Platform.

Prerequisites

- A valid Red Hat OpenShift Data Foundation Advanced subscription. For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.
- For Red Hat Enterprise Linux® operating system, ensure that there is iSCSI connectivity and then configure Linux multipath devices on the host.
- For Red Hat Enterprise Linux CoreOS or when the packages are already installed, configure Linux multipath devices on the host.
- Ensure to configure each worker with storage connectivity according to your storage system instructions. For the latest supported FlashSystem storage systems and versions, see IBM ODF FlashSystem driver documentation.
- Make sure that that the operator, **ibm-storage-odf-operator** is installed with the following subscription:

```
apiVersion: operators.coreos.com/v1alpha1
kind: Subscription
metadata:
  name: ibm-storage-odf-operator
  namespace: openshift-storage
spec:
  channel: stable-v1.8
  installPlanApproval: Automatic
  name: ibm-storage-odf-operator
  source: certified-operators
  sourceNamespace: openshift-marketplace
```

Procedure

1. In the OpenShift Web Console, click **Storage → External Systems**.
2. Click **Connect external systems**.
3. Select **IBM FlashSystem** and click **Next**.
4. In the Create storage class page, provide the following information:
   a. Enter the following details of IBM FlashSystem connection:
      - IP address
      - User name
      - Password
      - Pool name
   b. Select **thick** or **thin** for the **Volume mode**.
   c. Click **Next**.
5. In the Capacity and nodes page, provide the necessary details:
   a. Select a value for Requested capacity.
   The available options are **0.5 TiB**, **2 TiB**, and **4 TiB**. The requested capacity is dynamically allocated on the infrastructure storage class.
   b. Select at least three nodes in three different zones.
   It is recommended to start with at least 14 CPUs and 34 GiB of RAM per node. If the nodes selected do not match the OpenShift Data Foundation cluster requirement of an aggregated 30 CPUs and 72 GiB of RAM, a minimal cluster will be deployed. For minimum starting node requirements, see the Resource requirements section in the Planning guide.
   c. Click **Next**.
6. Optional: In the Security and network page, provide the necessary details:
   a. To enable encryption, select **Enable data encryption for block and file storage**
      i. Choose any one or both Encryption level:
         - **Cluster-wide encryption** to encrypt the entire cluster (block and file).
         - **StorageClass encryption** to create encrypted persistent volume (block only) using encryption enabled storage class.
      ii. Select the **Connect to an external key management service**checkbox. This is optional for cluster-wide encryption.
         A. Key Management Service Provider is set to Vault by default.
         B. Enter Vault Service Name, host Address of Vault server ('https://<hostname or ip>'), Port number, and Token.

iii. Expand Advanced Settings to enter additional settings and certificate details based on your Vault configuration:

A. Enter the Key Value secret path in the Backend Path that is dedicated and unique to OpenShift Data Foundation.

B. Optional: Enter TLS Server Name and Vault Enterprise Namespace.

C. Provide CA Certificate, Client Certificate, and Client Private Key by uploading the respective PEM encoded certificate file.

iv. Click **Save**.

v. Select **Default (SDN)** if you are using a single network or **Custom (Multus)** if you are using multiple network interfaces.

A. Select a Public Network Interface from the dropdown.

B. Select a Cluster Network Interface from the dropdown. NOTE: If you are using only one additional network interface, select the single **NetworkAttachementDefinition**, that is, **ocs-public-cluster** for the Public Network Interface, and leave the Cluster Network Interface blank.

vi. Click Next.

b. To enable in-transit encryption, select **In-transit encryption**.

i. Select a **Network**.

ii. Click **Next**.

7. In the Review and create page, review if all the details are correct:

- To modify any configuration settings, click **Back** to go back to the previous configuration page.

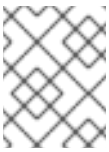
8. Click **Create StorageSystem**.

Verification

Verifying the state of the pods

1. Click **Workloads → Pods** from the left pane of the OpenShift Web Console.

2. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

Pods corresponding to OpenShift Data Foundation components

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any worker node) • **ocs-metrics-exporter-*** (1 pod on any worker node) • **odf-operator-controller-manager-*** (1 pod on any worker node) • **odf-console-*** (1 pod on any worker node) • **csi-addons-controller-manager-*** (1 pod on any worker node)
ibm-storage-odf-operator	• **ibm-storage-odf-operator-*** (2 pods on any worker nodes) • **ibm-odf-console-***
ibm-flashsystem-storage	**ibm-flashsystem-storage-*** (1 pod on any worker node)
rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any worker node)
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any worker node) • **noobaa-core-*** (1 pod on any worker node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)
CSI	• **ibm-block-csi-*** (1 pod on any worker node)

Verifying that the OpenShift Data Foundation cluster is healthy

1. In the Web Console, click **Storage → Data Foundation**.
2. In the Status card of the **Overview** tab, verify that the *Storage System* has a green tick mark.
3. In the **Details** card, verify that the cluster information is displayed.

For more information on the health of the OpenShift Data Foundation cluster using the Block and File dashboard, see Monitoring OpenShift Data Foundation.

Verifying that the Multicloud Object Gateway is healthy

1. In the Web Console, click **Storage → Data Foundation**.

2. In the Status card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up.

3. In the Status card of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.

4. In the **Details** card, verify that the MCG information is displayed.

For more information on the health of the OpenShift Data Foundation cluster using the object dashboard, see Monitoring OpenShift Data Foundation.

Verifying that IBM FlashSystem is connected and the storage cluster is ready

- Run the following command to verify if the OpenShift Data Foundation cluster is connected to the external IBM FlashSystem.

```
$ oc get flashsystemclusters.odf.ibm.com
NAME                              AGE   PHASE   CREATED AT
ibm-flashsystem-storage-fab3p-60   46h   Ready   2025-06-30T12:17:06Z
```

Verifying the subscription of the IBM operator

- Run the following command to verify the subscription:

```
$ oc get subscriptions.operators.coreos.com
NAME                                                                PACKAGE                SOURCE
CHANNEL
cephcsi-operator-stable-4.22-redhat-operators-openshift-marketplace         cephcsi-operator
redhat-operators          stable-4.22
ibm-block-csi-operator-stable-certified-operators-openshift-marketplace     ibm-block-csi-operator
certified-operators       stable
ibm-storage-odf-operator                                            ibm-storage-odf-operator   ibm-storage-
odf-catalog   stable-v1.7
mcg-operator-stable-4.22-redhat-operators-openshift-marketplace             mcg-operator
redhat-operators          stable-4.22
ocs-client-operator-stable-4.22-redhat-operators-openshift-marketplace      ocs-client-operator
redhat-operators          stable-4.22
ocs-operator-stable-4.22-redhat-operators-openshift-marketplace             ocs-operator
redhat-operators          stable-4.22
odf-csi-addons-operator-stable-4.22-redhat-operators-openshift-marketplace  odf-csi-addons-
operator    redhat-operators          stable-4.22
odf-dependencies                                                    odf-dependencies           redhat-operators
stable-4.22
odf-operator                                                        odf-operator               redhat-operators
stable-4.22
odf-prometheus-operator-stable-4.22-redhat-operators-openshift-marketplace  odf-prometheus-
operator    redhat-operators          stable-4.22
recipe-stable-4.22-redhat-operators-openshift-marketplace                   recipe                     redhat-
operators          stable-4.22
rook-ceph-operator-stable-4.22-redhat-operators-openshift-marketplace       rook-ceph-operator
redhat-operators          stable-4.22                                       odf-operator               odf-
catalogsource      alpha
```

Verifying the CSVs

- Run the following command to verify that the CSVs are in the succeeded state.

```
$ oc get csv
NAME                                  DISPLAY                           VERSION        REPLACES
PHASE
cephcsi-operator.v4.22-rhodf          CephCSI operator                  4.22-rhodf     cephcsi-
operator.v4.22-rhodf          Succeeded
ibm-block-csi-operator.v1.12.4        IBM block storage CSI driver operator   1.12.4
Succeeded
ibm-storage-odf-operator.v1.7.1       IBM Storage ODF operator          1.7.1          ibm-storage-
odf-operator.v1.7.0           Succeeded
mcg-operator.v4.22-rhodf              NooBaa Operator                   4.22-rhodf     mcg-
operator.v4.22-rhodf              Succeeded
ocs-client-operator.v4.22-rhodf       OpenShift Data Foundation Client  4.22-rhodf     ocs-client-
operator.v4.22-rhodf       Succeeded
ocs-operator.v4.22-rhodf              OpenShift Container Storage       4.22-rhodf     ocs-
operator.v4.22-rhodf              Succeeded
odf-csi-addons-operator.v4.22-rhodf   CSI Addons                        4.22-rhodf     odf-csi-addons-
operator.v4.22-rhodf   Succeeded
odf-dependencies.v4.22-rhodf          Data Foundation Dependencies      4.22-rhodf     odf-
dependencies.v4.22-rhodf          Succeeded
odf-operator.v4.22-rhodf              OpenShift Data Foundation         4.22-rhodf     odf-
operator.v4.22-rhodf              Succeeded
odf-prometheus-operator.v4.22-rhodf   Prometheus Operator               4.22-rhodf     odf-
prometheus-operator.v4.22-rhodf   Succeeded
recipe.v4.22-rhodf                    Recipe                            4.22-rhodf     recipe.v4.22-rhodf
Succeeded
rook-ceph-operator.v4.22-rhodf        Rook-Ceph
```

Verifying the IBM operator and CSI pods

- Run the following command to verify the IBM operator and CSI pods:

```
$ oc get pods
NAME                                                READY   STATUS             RESTARTS   AGE
ceph-csi-controller-manager-65f4bc4cd5-9qbp4        2/2     Running            0          46h
csi-addons-controller-manager-757d9c6748-94nrt      2/2     Running            0          46h
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-b642k                         3/3     Running
0          46h
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-l75pt                         3/3     Running
0          46h
openshift-storage.cephfs.csi.ceph.com-nodeplugin-847f7c54d8-52nz9              7/7     Running
0          46h
openshift-storage.cephfs.csi.ceph.com-nodeplugin-847f7c54d8-xtp5j              7/7     Running
0          46h
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-vh485                         3/3     Running
0          46h
openshift-storage.rbd.csi.ceph.com-ctrlplugin-9hss7                            4/4     Running
0          46h
openshift-storage.rbd.csi.ceph.com-ctrlplugin-9q9qd                            4/4     Running
0          46h
openshift-storage.rbd.csi.ceph.com-ctrlplugin-9vkqh                            4/4     Running
0          46h
openshift-storage.rbd.csi.ceph.com-nodeplugin-7f4b8d946f-ngz65                 7/7     Running
0          46h
```

```
openshift-storage.rbd.csi.ceph.com-nodeplugin-7f4b8d946f-x8l8t                    7/7     Running
0          46h
ibm-block-csi-controller-0                                        8/8     Running             0          41h
ibm-block-csi-node-6vmpq                                          3/3     Running             0          46h
ibm-block-csi-node-fw7xc                                          3/3     Running             0          41h
ibm-block-csi-node-k57sv                                          3/3     Running             0          46h
ibm-block-csi-operator-867c54df5d-2t968                           1/1     Running             0          46h
ibm-flashsystem-storage-deployment-5958ffd56b-tj6zw               1/1     Running             0
46h
ibm-odf-console-6d64dcc46c-bkknx                                  1/1     Running             0          41h
ibm-storage-odf-operator-787c67c7bf-nzp9w                         2/2     Running             0          41h
noobaa-core-0                                                     2/2     Running             0          46h
noobaa-db-pg-0                                                    1/1     Running             0          46h
noobaa-endpoint-5d94d888fc-bj28x                                  1/1     Running             0          46h
noobaa-operator-67b569fc76-rk99d                                  1/1     Running             0          46h
ocs-metrics-exporter-6f7f79fcf9-xm6pq                             2/3     Running             0          46h
ocs-operator-676c964c58-nsb7g                                     1/1     Running             0          46h
odf-console-75dc785696-n97sk                                      1/1     Running             0          46h
odf-operator-controller-manager-785ddcd999-tf245                  2/2     Running             0          46h
rook-ceph-crashcollector-worker-node1-b95b5cd54-7qbhb             1/1     Running             0
46h
rook-ceph-crashcollector-worker-node2-757d746d5f-jsfrx            1/1     Running             0
46h
rook-ceph-crashcollector-worker-node3-6d4f8f7848-xvvw2            1/1     Running             0
46h
rook-ceph-exporter-worker-node1-5dcf74b647-t5rg7                  1/1     Running             0
46h
rook-ceph-exporter-worker-node2-66c694f5b6-n96g5                  1/1     Running             0
46h
rook-ceph-exporter-worker-node3-58fd7ffcfc-vpzfr                  1/1     Running             0          46h
rook-ceph-mds-ocs-storagecluster-cephfilesystem-a-557b4cc9gsx65   2/2     Running             0
46h
rook-ceph-mds-ocs-storagecluster-cephfilesystem-b-5d8748dfmrhfn   2/2     Running             0
46h
rook-ceph-mgr-a-f56c6866b-tzl2s                                  3/3     Running             0          46h
rook-ceph-mgr-b-796d5f69bf-f54qr                                  3/3     Running             0          46h
rook-ceph-mon-a-6b5dc9dd6c-f4dvn                                  2/2     Running             0          46h
rook-ceph-mon-b-8d458fbdf-wjgzm                                   2/2     Running             0          46h
rook-ceph-mon-c-5b78749887-mcpjl                                  2/2     Running             0          46h
rook-ceph-operator-578dfdd8dd-mn9vn                               1/1     Running             0          46h
rook-ceph-osd-0-754499c4bd-2nrs2                                  2/2     Running             0          46h
rook-ceph-osd-1-5566ff6d99-dc6l4                                  2/2     Running             0          46h
rook-ceph-osd-2-8567c9f9f7-wr66b                                  2/2     Running             0          46h
rook-ceph-osd-prepare-667012a10f7f0800994523ce6731b5d1-n7fgb       0/1     Completed
0          46h
rook-ceph-osd-prepare-ab6be2d13c76aad7123b793996562bf0-wpqsv      0/1     Completed
0          46h
rook-ceph-osd-prepare-baf6cafc480b4e046c337c49601633b8-zllnx      0/1     Completed           0
46h
rook-ceph-rgw-ocs-storagecluster-cephobjectstore-a-5f4dfd9wlbvn   2/2     Running             0
46h
ux-backend-server-db4c5b7bd-hn9jx                                 2/2     Running             0          46h
```

CHAPTER 4. UNINSTALLING OPENSHIFT DATA FOUNDATION FROM EXTERNAL STORAGE SYSTEM

Use the steps in this section to uninstall OpenShift Data Foundation. Uninstalling OpenShift Data Foundation does not remove the RBD pool from the external cluster, or uninstall the external Red Hat Ceph Storage cluster.

Uninstall Annotations

Annotations on the Storage Cluster are used to change the behavior of the uninstall process. To define the uninstall behavior, the following two annotations have been introduced in the storage cluster:

- **uninstall.ocs.openshift.io/cleanup-policy: delete**
- **uninstall.ocs.openshift.io/mode: graceful**



NOTE

The **uninstall.ocs.openshift.io/cleanup-policy** is not applicable for external mode.

The below table provides information on the different values that can used with these annotations:

uninstall.ocs.openshift.io uninstall annotations descriptions

Annotation	Value	Default	Behavior
cleanup-policy	delete	Yes	Rook cleans up the physical drives and the **DataDirHostPath**
cleanup-policy	retain	No	Rook does **not** clean up the physical drives and the **DataDirHostPath**
mode	graceful	Yes	Rook and NooBaa **pauses** the uninstall process until the PVCs and the OBCs are removed by the administrator/user
mode	forced	No	Rook and NooBaa proceeds with uninstall even if PVCs/OBCs provisioned using Rook and NooBaa exist respectively

You can change the uninstall mode by editing the value of the annotation by using the following commands:

```
$ oc annotate storagecluster ocs-external-storagecluster -n openshift-storage
uninstall.ocs.openshift.io/mode="forced" --overwrite
storagecluster.ocs.openshift.io/ocs-external-storagecluster annotated
```

Prerequisites

- Ensure that the OpenShift Data Foundation cluster is in a healthy state. The uninstall process can fail when some of the pods are not terminated successfully due to insufficient resources or nodes. In case the cluster is in an unhealthy state, contact Red Hat Customer Support before uninstalling OpenShift Data Foundation.
- Ensure that applications are not consuming persistent volume claims (PVCs) or object bucket claims (OBCs) using the storage classes provided by OpenShift Data Foundation.

Procedure

1. Delete the volume snapshots that are using OpenShift Data Foundation.
   a. List the volume snapshots from all the namespaces

   ```
   $ oc get volumesnapshot --all-namespaces
   ```

   b. From the output of the previous command, identify and delete the volume snapshots that are using OpenShift Data Foundation.

   ```
   $ oc delete volumesnapshot <VOLUME-SNAPSHOT-NAME> -n <NAMESPACE>
   ```

2. Delete PVCs and OBCs that are using OpenShift Data Foundation.
   In the default uninstall mode (graceful), the uninstaller waits till all the PVCs and OBCs that use OpenShift Data Foundation are deleted.

   If you wish to delete the Storage Cluster without deleting the PVCs beforehand, you may set the uninstall mode annotation to "forced" and skip this step. Doing so will result in orphan PVCs and OBCs in the system.

   a. Delete OpenShift Container Platform monitoring stack PVCs using OpenShift Data Foundation.
   See Removing monitoring stack from OpenShift Data Foundation

   b. Delete OpenShift Container Platform Registry PVCs using OpenShift Data Foundation.
   Removing OpenShift Container Platform registry from OpenShift Data Foundation

   c. Delete OpenShift Container Platform logging PVCs using OpenShift Data Foundation.
   Removing the cluster logging operator from OpenShift Data Foundation

   d. Delete other PVCs and OBCs provisioned using OpenShift Data Foundation.

   - Given below is a sample script to identify the PVCs and OBCs provisioned using OpenShift Data Foundation. The script ignores the PVCs and OBCs that are used internally by OpenShift Data Foundation.

   ```
   #!/bin/bash

   RBD_PROVISIONER="openshift-storage.rbd.csi.ceph.com"
   CEPHFS_PROVISIONER="openshift-storage.cephfs.csi.ceph.com"
   ```

```
NOOBAA_PROVISIONER="openshift-storage.noobaa.io/obc"
RGW_PROVISIONER="openshift-storage.ceph.rook.io/bucket"

NOOBAA_DB_PG_1_PVC="noobaa-db-pg-cluster-1"
NOOBAA_DB_PG_2_PVC="noobaa-db-pg-cluster-2"
NOOBAA_BACKINGSTORE_PVC="noobaa-default-backing-store-noobaa-pvc"

# Find all the OCS StorageClasses
OCS_STORAGECLASSES=$(oc get storageclasses | grep -e
"$RBD_PROVISIONER" -e "$CEPHFS_PROVISIONER" -e
"$NOOBAA_PROVISIONER" -e "$RGW_PROVISIONER" | awk '{print $1}')

# List PVCs in each of the StorageClasses
for SC in $OCS_STORAGECLASSES
do
        echo
"============================================================================
=="
        echo "$SC StorageClass PVCs and OBCs"
        echo
"============================================================================
=="
        oc get pvc  --all-namespaces --no-headers 2>/dev/null | grep $SC | grep -v -e
"$NOOBAA_DB_PVC" -e "$NOOBAA_BACKINGSTORE_PVC"
        oc get obc  --all-namespaces --no-headers 2>/dev/null | grep $SC
        echo
done
```

- Delete the OBCs.

  ```
  $ oc delete obc <obc name> -n <project name>
  ```

- Delete the PVCs.

  ```
  $ oc delete pvc <pvc name> -n <project-name>
  ```

  Ensure that you have removed any custom backing stores, bucket classes, and so on that are created in the cluster.

3. Delete the Storage Cluster object and wait for the removal of the associated resources.

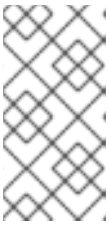
   ```
   $ oc delete -n openshift-storage storagecluster --all --wait=true
   ```

4. Delete the namespace and wait until the deletion is complete. You will need to switch to another project if **openshift-storage** is the active project.
   For example:

   ```
   $ oc project default
   $ oc delete project openshift-storage --wait=true --timeout=5m
   ```

   The project is deleted if the following command returns a **NotFound** error.

   ```
   $ oc get project openshift-storage
   ```



NOTE

While uninstalling OpenShift Data Foundation, if the namespace is not deleted completely and remains in **Terminating** state, perform the steps in Troubleshooting and deleting remaining resources during Uninstall to identify objects that are blocking the namespace from being terminated.

5. Confirm all PVs provisioned using OpenShift Data Foundation are deleted. If there is any PV left in the **Released** state, delete it.

```
$ oc get pv
$ oc delete pv <pv name>
```

6. Remove **CustomResourceDefinitions**.

```
$ oc delete crd backingstores.noobaa.io bucketclasses.noobaa.io
cephblockpools.ceph.rook.io cephclusters.ceph.rook.io cephfilesystems.ceph.rook.io
cephnfses.ceph.rook.io cephobjectstores.ceph.rook.io cephobjectstoreusers.ceph.rook.io
noobaas.noobaa.io ocsinitializations.ocs.openshift.io storageclusters.ocs.openshift.io
cephclients.ceph.rook.io cephobjectrealms.ceph.rook.io cephobjectzonegroups.ceph.rook.io
cephobjectzones.ceph.rook.io cephrbdmirrors.ceph.rook.io storagesystems.odf.openshift.io --
wait=true --timeout=5m
```

7. To ensure that OpenShift Data Foundation is uninstalled completely:

   a. In the OpenShift Container Platform Web Console, click **Storage**.

   b. Verify that **OpenShift Data Foundation** no longer appears under Storage.

4.1. REMOVING MONITORING STACK FROM OPENSHIFT DATA FOUNDATION

Remove the OpenShift monitoring stack from OpenShift Data Foundation external mode storage before uninstalling the deployment to reclaim the PVCs used by Prometheus and Alertmanager.

The PVCs that are created as a part of configuring the monitoring stack are in the **openshift-monitoring** namespace.

Prerequisites

- PVCs are configured to use the OpenShift Container Platform monitoring stack. For information, see configuring monitoring stack.

Procedure

1. List the pods and PVCs that are currently running in the **openshift-monitoring** namespace.

```
$ oc get pod,pvc -n openshift-monitoring
NAME                        READY   STATUS    RESTARTS   AGE
pod/alertmanager-main-0     3/3     Running   0          8d
pod/alertmanager-main-1     3/3     Running   0          8d
pod/alertmanager-main-2     3/3     Running   0          8d
pod/cluster-monitoring-
```

```
operator-84457656d-pkrxm        1/1     Running   0          8d
pod/grafana-79ccf6689f-2ll28    2/2     Running   0          8d
pod/kube-state-metrics-
7d86fb966-rvd9w                 3/3     Running   0          8d
pod/node-exporter-25894         2/2     Running   0          8d
pod/node-exporter-4dsd7         2/2     Running   0          8d
pod/node-exporter-6p4zc         2/2     Running   0          8d
pod/node-exporter-jbjvg         2/2     Running   0          8d
pod/node-exporter-jj4t5         2/2     Running   0          6d18h
pod/node-exporter-k856s         2/2     Running   0          6d18h
pod/node-exporter-rf8gn         2/2     Running   0          8d
pod/node-exporter-rmb5m         2/2     Running   0          6d18h
pod/node-exporter-zj7kx         2/2     Running   0          8d
pod/openshift-state-metrics-
59dbd4f654-4clng                3/3     Running   0          8d
pod/prometheus-adapter-
5df5865596-k8dzn                1/1     Running   0          7d23h
pod/prometheus-adapter-
5df5865596-n2gj9                1/1     Running   0          7d23h
pod/prometheus-k8s-0            6/6     Running   1          8d
pod/prometheus-k8s-1            6/6     Running   1          8d
pod/prometheus-operator-
55cfb858c9-c4zd9                1/1     Running   0          6d21h
pod/telemeter-client-
78fc8fc97d-2rgfp                3/3     Running   0          8d

NAME                                         STATUS   VOLUME
CAPACITY   ACCESS MODES   STORAGECLASS                AGE
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-0   Bound    pvc-0d519c4f-
15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-external-storagecluster-ceph-rbd
8d
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-1   Bound    pvc-
0d5a9825-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-external-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-2   Bound    pvc-
0d6413dc-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-external-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-prometheus-claim-prometheus-k8s-0        Bound    pvc-0b7c19b0-
15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-external-storagecluster-ceph-rbd
8d
persistentvolumeclaim/my-prometheus-claim-prometheus-k8s-1        Bound    pvc-0b8aed3f-
15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-external-storagecluster-ceph-rbd
8d
```

2. Edit the monitoring **configmap**.

```
$ oc -n openshift-monitoring edit configmap cluster-monitoring-config
```

Remove any **config** sections that reference the OpenShift Data Foundation storage classes as shown in the following example and save it.

Before editing

```
.
.
.
apiVersion: v1
data:
  config.yaml: |
    alertmanagerMain:
      volumeClaimTemplate:
        metadata:
          name: my-alertmanager-claim
        spec:
          resources:
            requests:
              storage: 40Gi
          storageClassName: ocs-external-storagecluster-ceph-rbd
    prometheusK8s:
      volumeClaimTemplate:
        metadata:
          name: my-prometheus-claim
        spec:
          resources:
            requests:
              storage: 40Gi
          storageClassName: ocs-external-storagecluster-ceph-rbd
kind: ConfigMap
metadata:
  creationTimestamp: "2019-12-02T07:47:29Z"
  name: cluster-monitoring-config
  namespace: openshift-monitoring
  resourceVersion: "22110"
  selfLink: /api/v1/namespaces/openshift-monitoring/configmaps/cluster-monitoring-config
  uid: fd6d988b-14d7-11ea-84ff-066035b9efa8


.
.
.
```

After editing

```
.
.
.
apiVersion: v1
data:
  config.yaml: |
kind: ConfigMap
metadata:
  creationTimestamp: "2019-11-21T13:07:05Z"
  name: cluster-monitoring-config
  namespace: openshift-monitoring
  resourceVersion: "404352"
  selfLink: /api/v1/namespaces/openshift-monitoring/configmaps/cluster-monitoring-config
  uid: d12c796a-0c5f-11ea-9832-063cd735b81c
.
.
.
```

In this example, **alertmanagerMain** and **prometheusK8s** monitoring components are using the OpenShift Data Foundation PVCs.

3. List the pods consuming the PVC.
   In this example, the **alertmanagerMain** and **prometheusK8s** pods that were consuming the PVCs are in the **Terminating** state. You can delete the PVCs once these pods are no longer using OpenShift Data Foundation PVC.

```
$ oc get pod,pvc -n openshift-monitoring
NAME                                          READY   STATUS      RESTARTS AGE
pod/alertmanager-main-0                       3/3     Terminating   0      10h
pod/alertmanager-main-1                       3/3     Terminating   0      10h
pod/alertmanager-main-2                       3/3     Terminating   0      10h
pod/cluster-monitoring-operator-84cd9df668-zhjfn   1/1     Running       0      18h
pod/grafana-5db6fd97f8-pmtbf                  2/2     Running       0      10h
pod/kube-state-metrics-895899678-z2r9q        3/3     Running       0      10h
pod/node-exporter-4njxv                       2/2     Running       0      18h
pod/node-exporter-b8ckz                       2/2     Running       0      11h
pod/node-exporter-c2vp5                       2/2     Running       0      18h
pod/node-exporter-cq65n                       2/2     Running       0      18h
pod/node-exporter-f5sm7                       2/2     Running       0      11h
pod/node-exporter-f852c                       2/2     Running       0      18h
pod/node-exporter-l9zn7                       2/2     Running       0      11h
pod/node-exporter-ngbs8                       2/2     Running       0      18h
pod/node-exporter-rv4v9                       2/2     Running       0      18h
pod/openshift-state-metrics-77d5f699d8-69q5x      3/3     Running       0      10h
pod/prometheus-adapter-765465b56-4tbxx        1/1     Running       0      10h
pod/prometheus-adapter-765465b56-s2qg2        1/1     Running       0      10h
pod/prometheus-k8s-0                          6/6     Terminating   1      9m47s
pod/prometheus-k8s-1                          6/6     Terminating   1      9m47s
pod/prometheus-operator-cbfd89f9-ldnwc        1/1     Running       0      43m
pod/telemeter-client-7b5ddb4489-2xfpz         3/3     Running       0      10h

NAME                                        STATUS  VOLUME
CAPACITY   ACCESS MODES   STORAGECLASS               AGE
```

```
persistentvolumeclaim/ocs-alertmanager-claim-alertmanager-main-0 Bound pvc-2eb79797-1fed-11ea-93e1-0a88476a6a64 40Gi RWO ocs-external-storagecluster-ceph-rbd 19h
persistentvolumeclaim/ocs-alertmanager-claim-alertmanager-main-1 Bound pvc-2ebeee54-1fed-11ea-93e1-0a88476a6a64 40Gi RWO ocs-external-storagecluster-ceph-rbd 19h
persistentvolumeclaim/ocs-alertmanager-claim-alertmanager-main-2 Bound pvc-2ec6a9cf-1fed-11ea-93e1-0a88476a6a64 40Gi RWO ocs-external-storagecluster-ceph-rbd 19h
persistentvolumeclaim/ocs-prometheus-claim-prometheus-k8s-0 Bound pvc-3162a80c-1fed-11ea-93e1-0a88476a6a64 40Gi RWO ocs-external-storagecluster-ceph-rbd 19h
persistentvolumeclaim/ocs-prometheus-claim-prometheus-k8s-1 Bound pvc-316e99e2-1fed-11ea-93e1-0a88476a6a64 40Gi RWO ocs-external-storagecluster-ceph-rbd 19h
```

4. Delete relevant PVCs. Make sure you delete all the PVCs that are consuming the storage classes.

```
$ oc delete -n openshift-monitoring pvc <pvc-name> --wait=true --timeout=5m
```

4.2. REMOVING OPENSHIFT CONTAINER PLATFORM REGISTRY FROM OPENSHIFT DATA FOUNDATION

Use this section to clean up the OpenShift Container Platform registry from OpenShift Data Foundation. If you want to configure an alternative storage, see image registry

The PVCs that are created as a part of configuring OpenShift Container Platform registry are in the **openshift-image-registry** namespace.

Prerequisites

- The image registry should have been configured to use an OpenShift Data Foundation PVC.

Procedure

1. Edit the **configs.imageregistry.operator.openshift.io** object and remove the content in the **storage** section.

```
$ oc edit configs.imageregistry.operator.openshift.io
```

Before editing

```
.
.
.
storage:
  pvc:
    claim: registry-cephfs-rwx-pvc
.
.
.
```

After editing

```
.
.
.
storage:
  emptyDir: {}
.
.
.
```

In this example, the PVC is called **registry-cephfs-rwx-pvc**, which is now safe to delete.

2. Delete the PVC.

```
$ oc delete pvc <pvc-name> -n openshift-image-registry --wait=true --timeout=5m
```

4.3. REMOVING THE CLUSTER LOGGING OPERATOR FROM OPENSHIFT DATA FOUNDATION

Use this section to clean up the cluster logging operator from OpenShift Data Foundation.

The Persistent Volume Claims (PVCs) that are created as a part of configuring the cluster logging operator are in the **openshift-logging** namespace.

Prerequisites

- The cluster logging instance should have been configured to use the OpenShift Data Foundation PVCs.

Procedure

1. Remove the **ClusterLogging** instance in the namespace.

```
$ oc delete clusterlogging instance -n openshift-logging --wait=true --timeout=5m
```

The PVCs in the **openshift-logging** namespace are now safe to delete.

2. Delete the PVCs.

```
$ oc delete pvc <pvc-name> -n openshift-logging --wait=true --timeout=5m
```

<pvc-name>

Is the name of the PVC

4.4. REMOVING EXTERNAL IBM FLASHSYSTEM SECRET

You need to clean up the FlashSystem secret from OpenShift Data Foundation while uninstalling. This secret is created when you configure the external IBM FlashSystem Storage. For more information, see Creating an OpenShift Data Foundation Cluster for external IBM FlashSystem storage .

Procedure

- Remove the IBM FlashSystem secret by using the following command:

```
$ oc delete secret -n openshift-storage ibm-flashsystem-storage
```